

IML 2025 Assignment 2

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- **Explanation of strategies/algorithms**

We implemented both the epsilon-greedy and Boltzmann exploration strategies, using Q-learning as the fixed learning algorithm to compare their performance.

For epsilon greedy exploration, we used epsilon decay starting from 1 and decaying to 0.01 exponentially by hyperparameter decay rate.

For Boltzmann exploration, we used softmax function to Q-values of each action to calculate pseudo-probabilities for action sampling. we used temperature decay starting from 1 and decaying to 0.01 exponentially by hyperparameter decay rate, which is different with epsilon greedy's decay rate.

We fixed our learning algorithm to Q-learning to compare the performance of two exploration strategies.

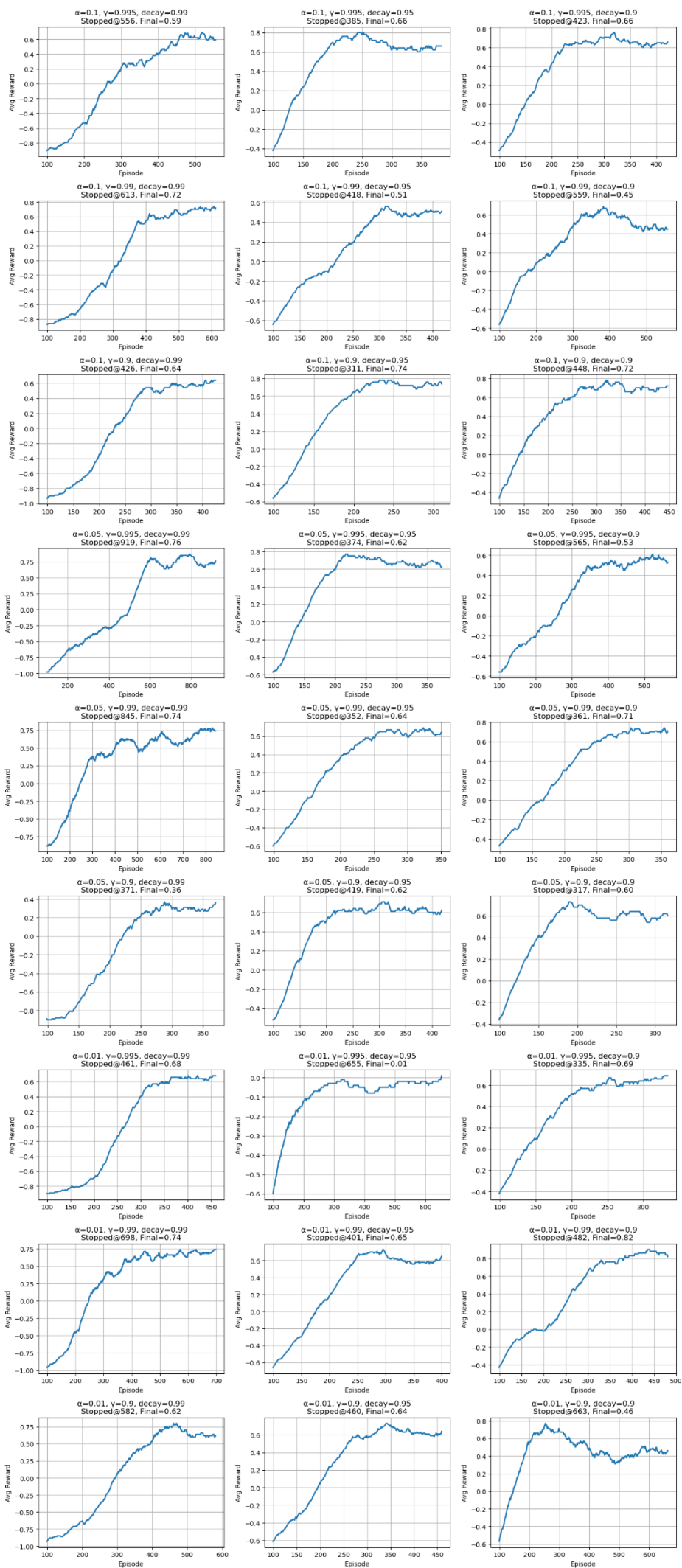
We used various hyperparameters, such as learning rate, decay rate, discount factor, to find optimal agent for each given strategy.

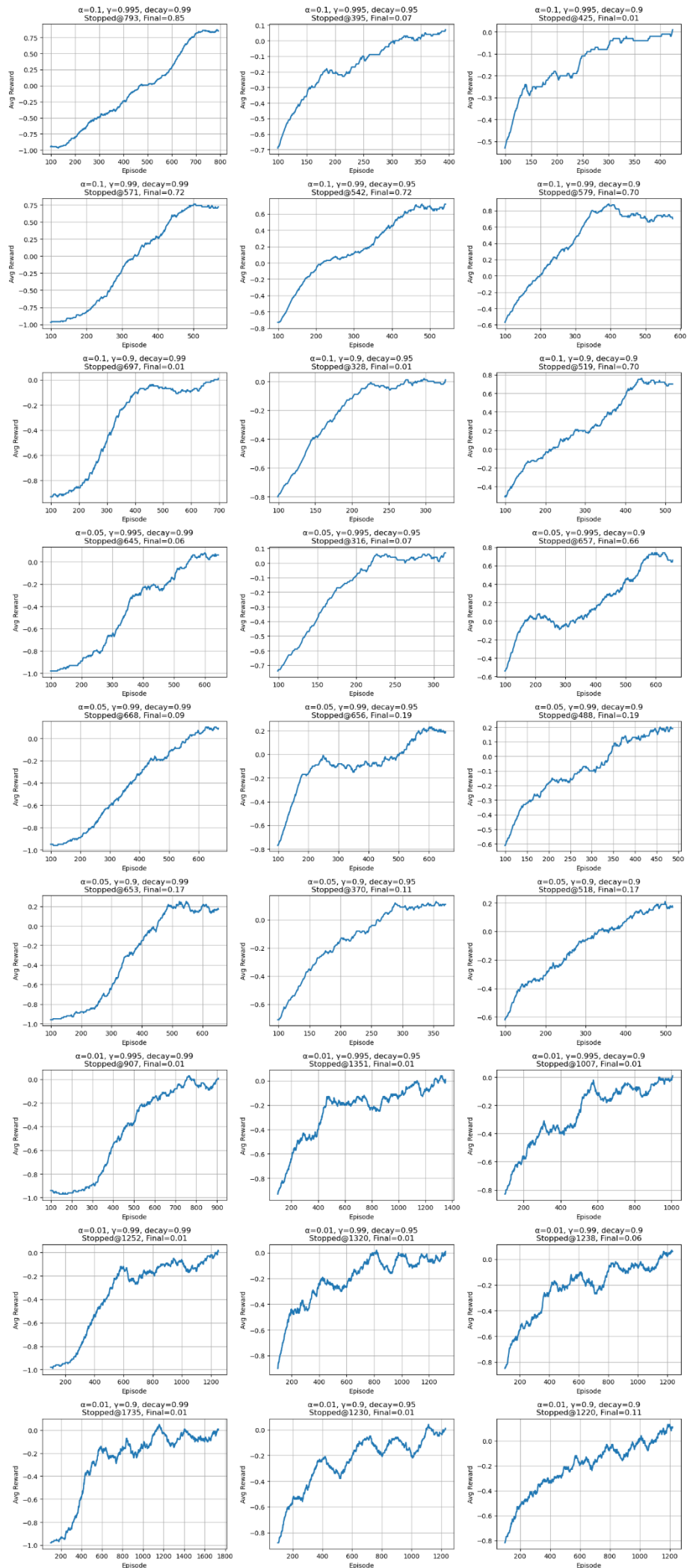
Also we have engineered rewards to boost the learning. we gave reward -1 if the agent falls into a hole. otherwise it is same.

- **Plots comparing the performance**

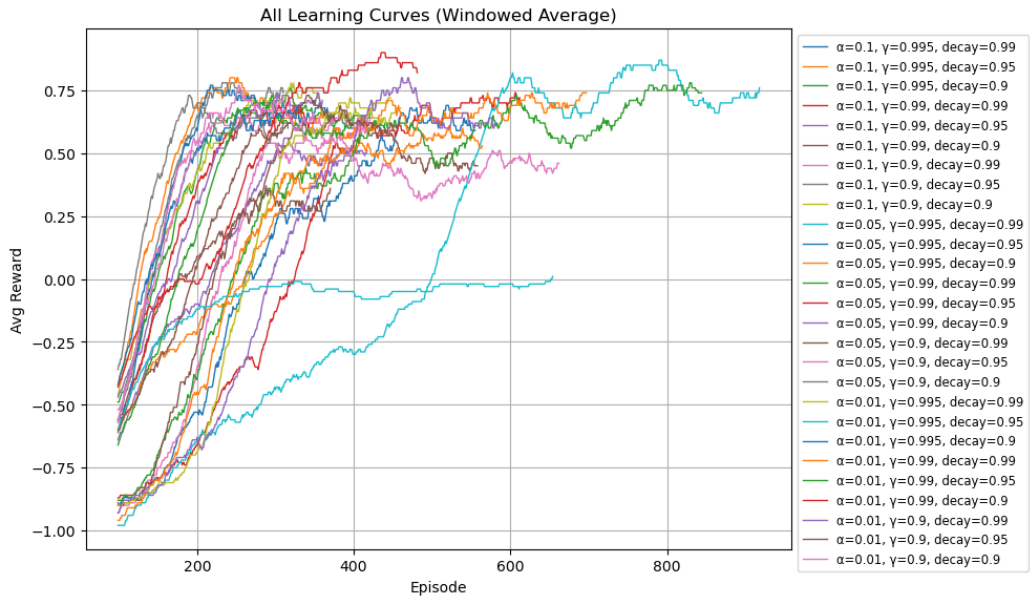
We plotted learning curve(number of episode vs cumulative reward) for each exploration strategy and hyperparameter set.

Cumulative reward has been calculated on window size 100, and early stopped if the difference of cumulative reward on specific time range is smaller than stopping tolerance.

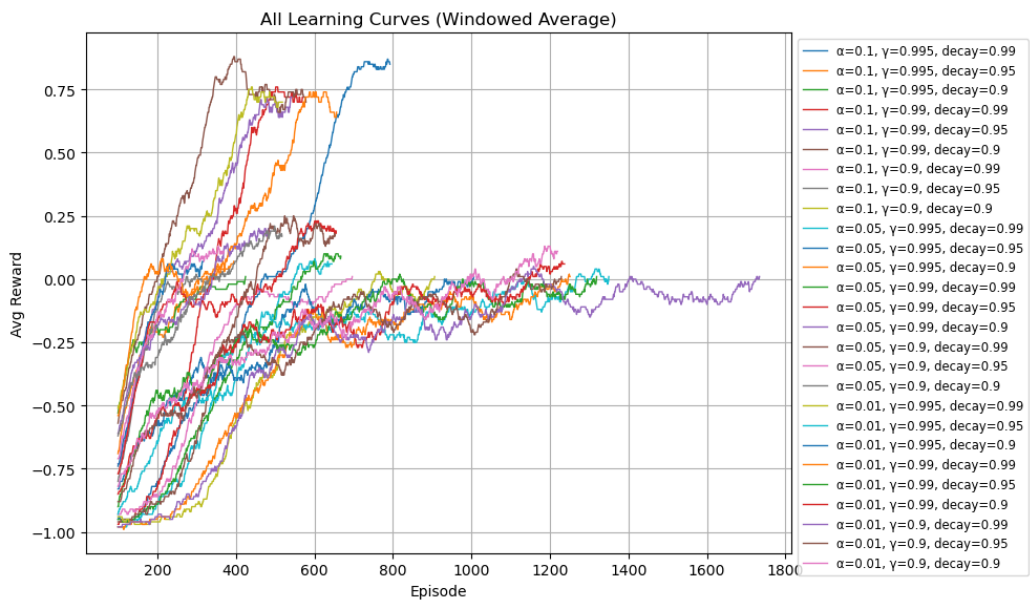




We merged all results of hyperparameter sets on two plots.

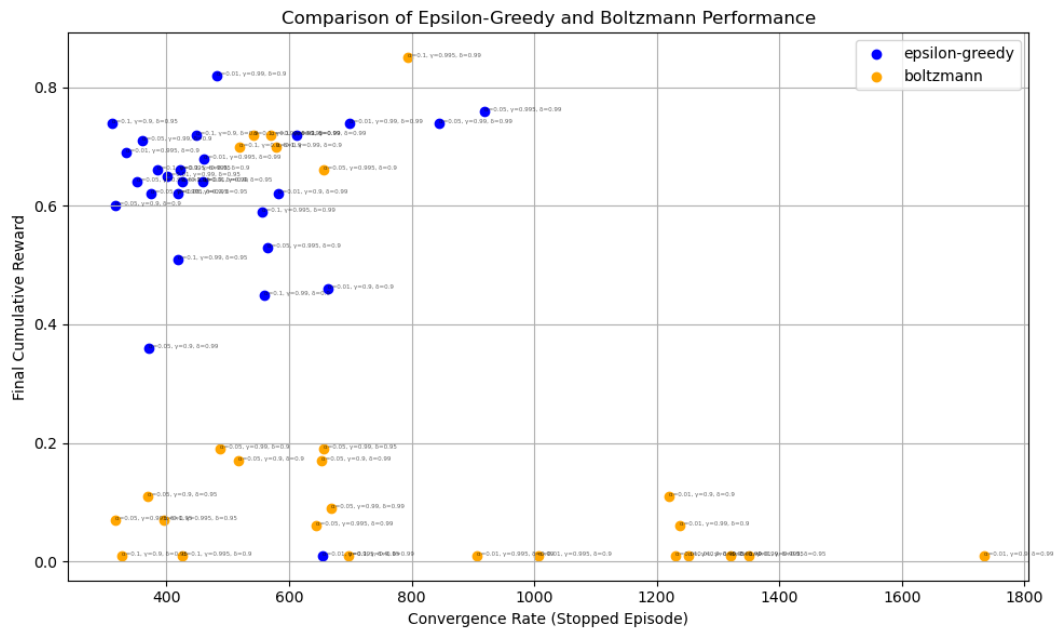


Learning Curves of epsilon-greedy exploration

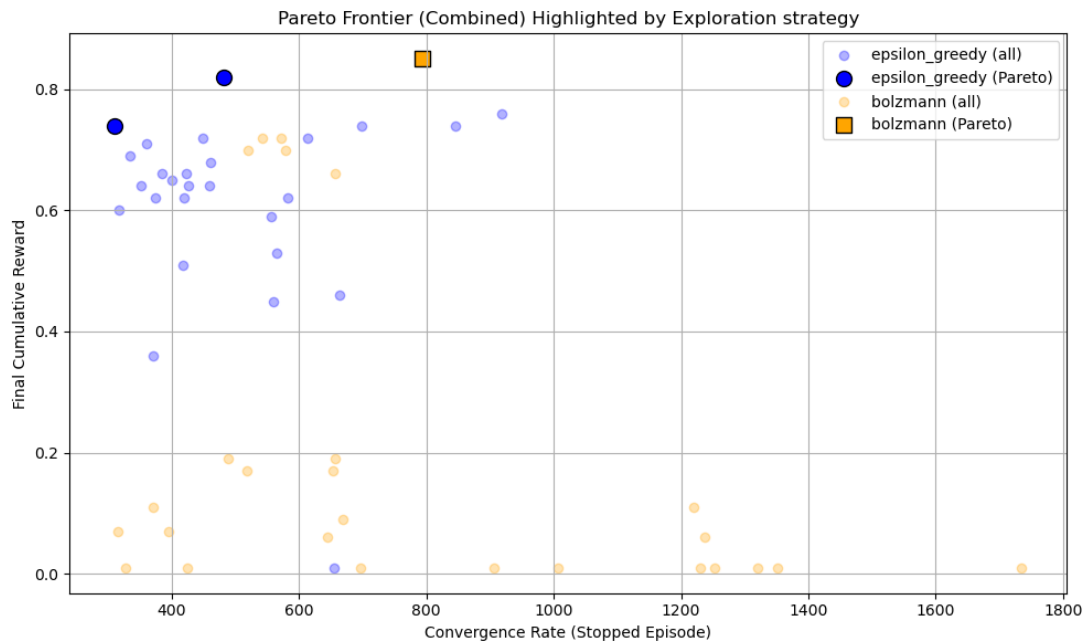


Learning Curves of boltzmann exploration

And the following plot is the scatter performance plot(convergence rate vs cumulative reward) of .two exploration strategies



The convergence rate and cumulative reward showed **weak trade-off** relationship, so we marked pareto frontiers of strategy-hyperparameter set to find the sub optimal models.



We got 2 epsilon-greedy pareto frontiers and 1 boltzmann frontiers.

The following table is the (sub)optimal hyperparameter-strategy sets.

Strategy	hyperparameters	Cumulative Reward	Convergence Rate
epsilon_greedy	$\alpha = 0.1, \gamma = 0.9,$ decay rate = 0.95	0.74	331
epsilon_greedy	$\alpha = 0.01, \gamma = 0.99,$ decay rate = 0.9	0.82	482
boltzmann	$\alpha = 0.1, \gamma = 0.995,$ decay rate = 0.99	0.85	793

- **Analysis of when and why the strategies/algorithms succeed or fail**

When the learning rate for update Q-table was low, most of the model showed low cumulative reward(Boltzmann exploration) or high convergence rate(low speed of convergence; epsilon greedy exploration), which can be improved by adjusting learning rate.

For Boltzmann exploration, when the learning rate and temperature decay rate were high (slowly decaying), the model gave cumulative reward.

For Epsilon-greedy exploration, when the epsilon decay rate was high, the model weakly tended to give good cumulative reward.

Based on the results, it was observed that a higher decay rate, meaning exploration is maintained for a longer period, weakly leads to higher cumulative rewards and indicates successful learning.

- **Which strategy would be picked if sample efficiency were critical?**

The **epsilon greedy would be more sample-efficient** because the pareto frontier points of epsilon greedy had much smaller converge rate but had decent cumulative reward.